ECON 2 Practice final C

Name: _____ SID: _____ · 90 minutes · 60 points · one page of notes, front and back · calculator allowed

The exam

Work on paper. Every graphing part shows the axes you would get printed on the exam, so copy them or print this page and draw on them. Each answer below is longer than you need so you can learn from it; the "Full credit" line says what a grader looks for. On the real exam he asks for answers inside a box and says "less is more," so aim for two to four sentences per part.

Rough time plan at about 1.5 minutes per point: 3 minutes on the cover page, 15 on Question 1, 15 on Question 2, 35 on Question 3, and 18 on Question 4, which leaves a few minutes to check your arithmetic.

Cover page (2 points)

2 points (1 point) Write your name and student ID at the top of each page of the exam. (1 point) If you could hand one Berkeley dining hall the job of running monetary policy, which one would you pick? (There are no wrong answers.)

Question 1: Multiple choice, maximum marginal benefit (12 points)

For each of these questions, circle the one correct answer.

2 points 1(a). A monopolist sells 10 units a day at a price of $20 each. To sell an 11th unit each day, it must lower its price to $19 on every unit it sells. What is the marginal revenue from the 11th unit?

(i) $19, the price of the 11th unit
(ii) $9
(iii) $20
(iv) $209

2 points 1(b). In one hour of work, a worker in Country A can produce 6 laptops or 3 tons of wheat. A worker in Country B can produce 2 laptops or 2 tons of wheat. Which country has the comparative advantage in wheat?

(i) Country A, because it produces more wheat per hour
(ii) Country A, because it has an absolute advantage in both goods
(iii) Country B, because its opportunity cost of a ton of wheat is 1 laptop, compared with 2 laptops in Country A
(iv) Neither country, so Country A should produce both goods

2 points 1(c). The Fed's target inflation rate is 2%, the normal real interest rate is 3%, and the Fed follows the reaction function $r - 3\% = 0.5 \times (\pi - 2\%)$, where r is the real interest rate and π is inflation. Inflation is currently 4%. What does the Fed do?

(i) Lowers the real interest rate to 2%, so the nominal interest rate is 6%
(ii) Keeps the real interest rate at 3%, so the nominal interest rate is 7%
(iii) Raises the real interest rate to 4%, so the nominal interest rate is 8%
(iv) Raises the real interest rate to 4%, so the nominal interest rate is 4%

2 points 1(d). A large group of new workers arrives in the labor market for farm work. Crop prices, farm equipment, and farming methods do not change. In this labor market,

(i) Labor supply shifts out, the wage falls, and employment rises
(ii) Labor demand shifts out, and both the wage and employment rise
(iii) Labor supply shifts out, and both the wage and employment fall
(iv) Nothing changes, because the marginal revenue product of labor did not change

2 points 1(e). Which of these is counted in U.S. gross domestic product (GDP) for 2026?

(i) A painting from 1890 that one collector sells directly to another in 2026
(ii) A U.S. government bond that a retiree buys in 2026
(iii) Unemployment benefits paid by state governments in 2026
(iv) A new car factory built in Tennessee in 2026 by a Japanese-owned company

2 points 1(f). When one person gets a flu shot, the people around that person also become less likely to catch the flu, and nobody pays the vaccinated person for that benefit. Without government intervention, the market for flu shots produces

(i) Too little, because social marginal benefit exceeds social marginal cost at the market quantity
(ii) Too much, because social marginal cost exceeds social marginal benefit at the market quantity
(iii) The efficient amount, because buyers set their own marginal benefit equal to the price
(iv) The efficient amount, because the extra benefit goes to other people and not to buyers

Question 2: Short answers, compounded interest in you (10 points)

Two points each. Show your work wherever there is a number.

2 points 2(a). A hypothetical country has real GDP per person of $5,000 today, and it grows 3.5% per year. Use the rule of 70 to find about how many years it takes for real GDP per person to reach $40,000. How many years would it take if growth were only 1.4% per year? Show your work.

2 points 2(b). A bond pays $50 one year from now and $1,050 two years from now. Find its price when the market interest rate is 5%. Then the Fed cuts rates and the market interest rate falls to 4%. What is the bond's new price, and why did it move that way? Show your work.

2 points 2(c). Suppose a campus job paid $16 per hour in 2016, when the consumer price index (CPI) was 240. The same job pays $21 per hour in 2026, when the CPI is 300. (These are made-up round numbers.) Did the real wage for this job rise, fall, or stay the same? By what percent? Show your work.

2 points 2(d). Two hypothetical neighboring countries have the same natural resources, the same climate, and access to the same technology. In the first, courts enforce contracts and the government rarely seizes property. In the second, officials often demand bribes and sometimes take over successful businesses. Which country will likely have higher average labor productivity in the long run? Explain using the determinants of productivity.

2 points 2(e). A U.S.-made laptop sells for $1,000 and a Japanese game console sells for ¥28,000. The exchange rate moves from ¥140 per $1 to ¥160 per $1. Did the dollar appreciate or depreciate? What happens to the yen price of the laptop, the dollar price of the console, and U.S. net exports? Show your work.

Question 3: Too much of a good thing in Pacifica (24 points)

Pacifica is a hypothetical economy whose potential output is Y* = $40 trillion. Its marginal propensity to consume (mpc) is 0.75, and this system describes planned aggregate expenditure (PAE) in trillions of dollars, where T is taxes:

C = 6 + 0.75(Y − T) T = 4 I = 4 G = 4 NX = −1 PAE = C + I + G + NX

2 points 3(a). Combine the expenditure terms in PAE and find the level of PAE when Y = 0, which is the vertical intercept in the Keynesian cross diagram. Show your work.

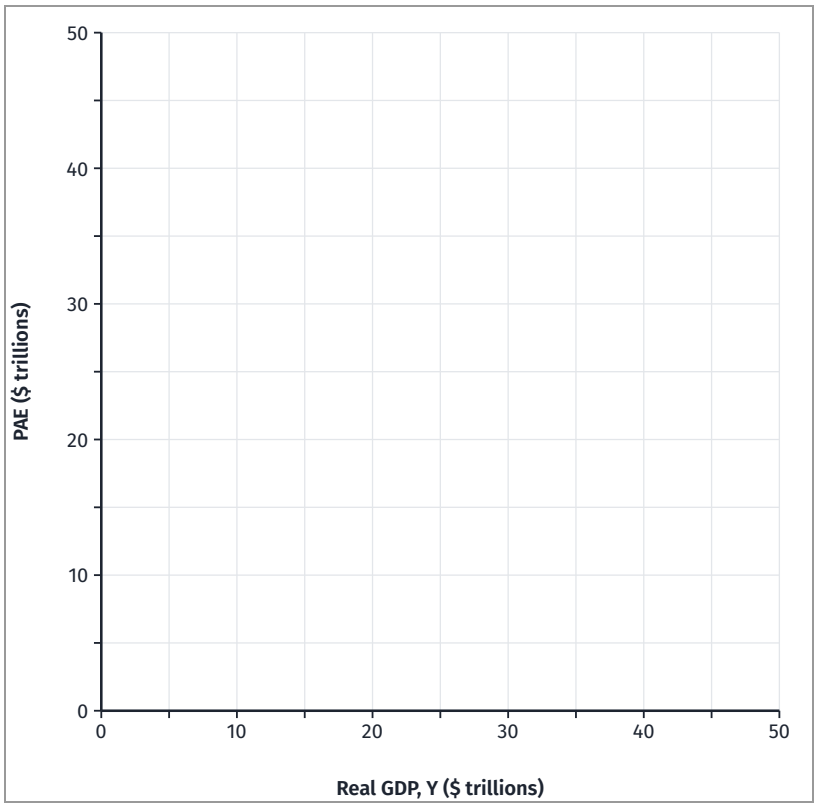
2 points 3(b). Set PAE = Y and solve for short-run equilibrium output. Is Pacifica at, above, or below potential? Show your work.

In 2027, a wave of new factory construction, as firms move production back to Pacifica, raises planned investment from 4 to I' = 5.5, and Congress passes a large infrastructure bill that raises government purchases from 4 to G' = 4.5. Nothing else changes.

2 points 3(c). Find the new expenditure line PAE' and its vertical intercept. Show your work.

2 points 3(d). Set PAE' = Y and solve for the new short-run equilibrium output. Show your work.

4 points 3(e). On the axes below, graph the original PAE line from (a), the new PAE' line from (c), and the 45-degree line. Show both equilibria, label the intercepts, and mark potential output Y* and the output gap.



Your printed axes for 3(e). Both axes use the same scale, so the 45-degree line passes through the grid corners (10, 10), (20, 20), and so on.

2 points 3(f). By how much did PAE change (at a given level of Y), and by how much did GDP change? Explain fully why the two numbers differ, and compute the multiplier.

2 points 3(g). What is Pacifica's output gap after the boom? Is it a recessionary or an expansionary gap? What would you expect to see in Pacifica's labor market?

2 points 3(h). Suppose Congress wants to bring output back to potential using government purchases alone. By how much must G change, and in which direction? What will G be afterward? Explain fully.

2 points 3(i). Suppose Congress does nothing. What happens to inflation in Pacifica right away, and what happens to it over the following year or two? Explain fully.

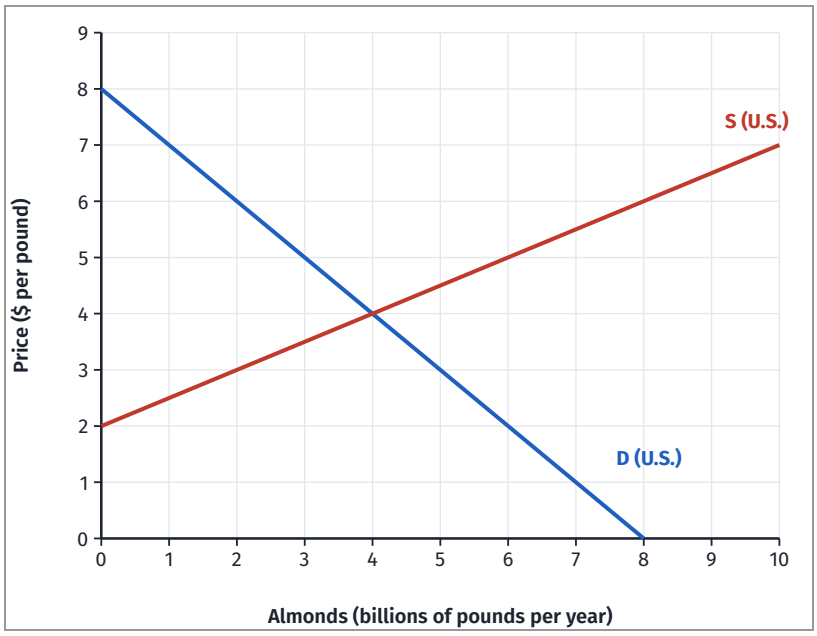
2 points 3(j). The Fed follows a reaction function. Explain fully how the Fed's response brings output back to potential, and sketch the shift on your graph from 3(e).

2 points 3(k). Suppose each 1 percentage point increase in the real interest rate lowers autonomous planned investment and consumption by a combined $1 trillion at every level of Y. By how many percentage points must the Fed raise r, in total, to return Pacifica to potential? In the long run, which kinds of spending are larger and which are smaller than before the boom?

Question 4: Almond joy, almond annoy (12 points)

Suppose the U.S. exports almonds to Europe, where they sell at a world price of €4.50 per pound. U.S. domestic demand for almonds is $P = 8 - Q$ and U.S. supply is $P = 2 + 0.5Q$, where Q is billions of pounds per year and P is dollars per pound. All numbers are hypothetical. The exchange rate starts at €0.75 per $1.

4 points 4(a). The diagram below shows the U.S. market. First find the price and quantity with no trade. Then convert the world price into dollars, and adapt the diagram to show the world price, U.S. quantity demanded, U.S. quantity supplied, and exports. Briefly explain what you drew.



Your printed diagram for 4(a): U.S. demand and supply for almonds.

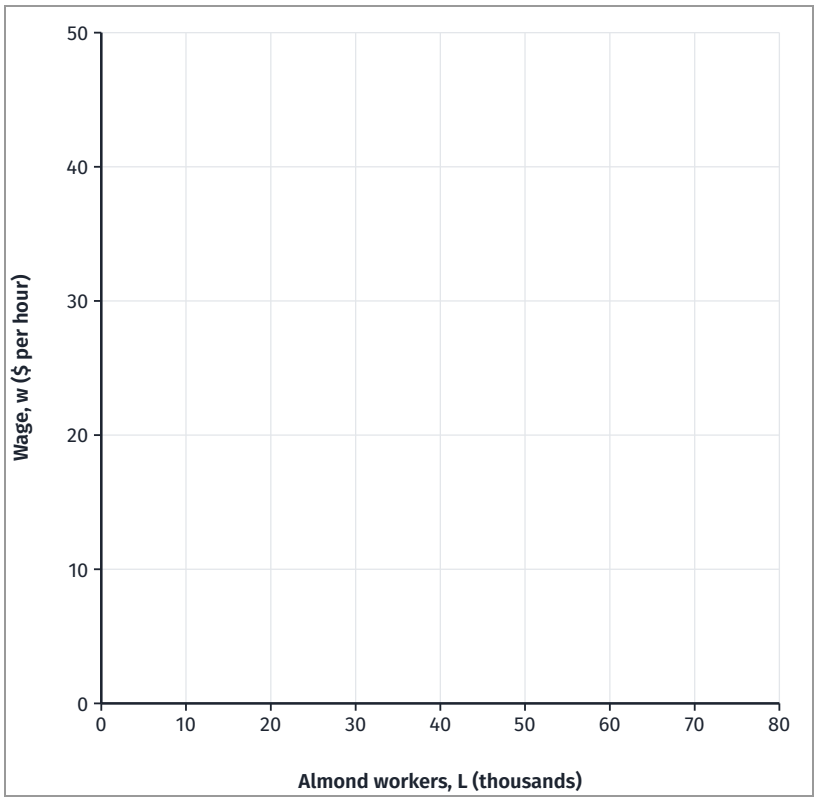
The Fed raises interest rates and global investors seek out U.S. assets, so the dollar appreciates from €0.75 to €0.90 per $1. The world price in Europe stays at €4.50 per pound.

2 points 4(b). What price in dollars do U.S. almond growers now receive? What happens to U.S. quantity demanded, quantity supplied, and exports? Show the new price on your graph from 4(a).

2 points 4(c). Calculate the change in U.S. consumer surplus, U.S. producer surplus, and total U.S. surplus. Who wins and who loses? Show your work.

Now consider the labor market for almond workers. Labor demand is the marginal revenue product of labor, $MRPL = MR \times MPL$, where MR is the price growers receive per pound and the marginal product of labor is $MPL = 8 - 0.1L$ pounds per worker-hour, with L in thousands of workers. Labor supply is $w = 4 + 0.5L$, where w is the hourly wage.

4 points 4(d). Find the equilibrium wage and employment before the appreciation (price $6) and after it (price $5). Illustrate both on the axes below, showing all demand and supply curves and both equilibria, and briefly explain.



Your printed axes for 4(d).